

OPERATING SYSTEM

Elements of a Computer System

- HARDWARE
- SOFTWARE
- PEOPLEWARE
- DATA

What is Operating System?

An Operating System (OS) is an interface between a computer user and computer hardware.

An Operating System (OS) is a collection of software that manages computer hardware resources and provides common services for computer programs.

Andy Rubin – founder of Android

Bill Gates – Microsoft

Operating System History Timeline

1. MS-DOS (Jul 1, 1981)

- Command-line only
- No mouse
- No graphics (GUI)
- Users type commands
- *MICROSOFT DISK OPERATING SYSTEM*
- Text-based

2. Windows 1.0 (Nov 20, 1985)

- First Windows GUI
- Mouse support
- Windows interface
- First graphical Windows
- *INTERFACE MANAGER* – Original name before "Windows"

3. Windows 2.0 (Dec 9, 1987)

- Overlapping windows
- Resize windows
- Better multitasking

4. Windows 3.0 (May 22, 1990)

- Better graphics
- Program Manager
- Icons
- *Solitaire* introduced
- User-friendly

5. Linux (Sep 17, 1991)

- Open source
- Free
- Created by *Linus Benedict Torvalds*

6. Windows 95 (Aug 24, 1995)

- *Start Button*
- Start Menu
- Taskbar
- Plug and Play

- Codename: *Chicago*
- First modern Windows interface

7. Windows 98 (Jun 25, 1998)

- USB support
- *Better Internet support*
- *More hardware compatibility*

8. Windows 2000 (Feb 17, 2000)

- Stable
- *Business OS*
- NT-based operating system

9. Windows ME (Jun 14, 2000)

- *System Restore*
- *MILLENNIUM EDITION*
- *Personal/home OS*
- Known for being *buggy*
- Nicknamed "*Mistake Edition*"

10. Mac OS X Public Beta (Sep 13, 2000)

- First public preview of Mac OS X
- Introduced the Aqua interface

11. Mac OS X 10.0 Cheetah (Mar 24, 2001)

- First official Mac OS X
- Aqua user interface
- UNIX-based system
- *Animal naming*
- *Very slow*

12. Mac OS X 10.1 Puma (Sep 25, 2001)

- Faster than Cheetah
- Better performance

13. Windows XP (Oct 25, 2001)

- Colorful interface
- Fast
- Stable
- *Very popular (14 years)*
- *User-friendly*
- XP = eXPerience

14. Mac OS X 10.2 Jaguar (Aug 23, 2002)

- Better speed
- Better usability
- Improved graphics
- *ichat*

15. Mac OS X 10.3 Panther (Oct 24, 2003)

- Faster Finder
- Exposé window management
- *Business friendly*

16. Mac OS X 10.4 Tiger (Apr 29, 2005)

- Spotlight Search
- Dashboard widgets
- *Long ver*
- *User-friendly*

17. Windows Vista (Jan 30, 2007)

- Aero Glass interface
- User Account Control (UAC)

- *Better security (Cap prob worst due to security issues and bugged)*

18. Mac OS X 10.5 Leopard (Oct 26, 2007)

- *Time Machine*
- Spaces (Virtual Desktops)
- First macOS with Time Machine backup
- *Restore system*

19. Windows 7 (Jul 22, 2009)

- Faster than Vista
- Better taskbar
- *More stable*
- Improved performance
- *Touchscreen*

20. Mac OS X 10.6 Snow Leopard (Aug 28, 2009)

- *Performance improvements*
- *Stability improvements*
- Few visual changes
- Better than Leopard

21. Mac OS X 10.7 Lion (Jul 20, 2011)

- Launchpad
- Full-screen apps
- Multi-touch gestures
- *Similar to iPad*

22. Mac OS X 10.8 Mountain Lion (Jul 25, 2012)

- Notification Center
- Messages
- Better iPhone/iPad integration
- *Anti-virus*

23. Windows 8 (Oct 26, 2012)

- *Metro UI*
- *Touchscreen interface*
- *Removed Start Menu*
- Live Tiles

24. Mac OS X 10.9 Mavericks (Jun 10, 2013)

- Better battery life
- Finder tabs
- *First version named after a California location*
- Free OS upgrades
- *Broke cat naming*

25. Mac OS X 10.10 Yosemite (Oct 16, 2014)

- Major redesign (*aesthetic*)
- Flat interface
- Continuity with iPhone

26. Windows 10 (Jul 29, 2015)

- *Start Menu returns*
- Virtual desktops
- Microsoft Edge
- Windows as a Service
- *Windows 9 skipped (legacy code issue)*

27. Mac OS X 10.11 El Capitan (Sep 30, 2015)

- *Split View*
- Faster performance
- Better multitasking

28. macOS X 10.12 Sierra (Sep 20, 2016)

- *Siri on Mac*
- Apple Pay
- Universal Clipboard

29. macOS X 10.13 High Sierra (Sep 25, 2017)

- APFS (Apple File System)
- Better graphics
- Faster storage
- *Better performance than Sierra*

30. macOS X 10.14 Mojave (Sep 24, 2018)

- *Dark Mode*
- Dynamic Desktop
- Desktop Stacks

31. macOS X 10.15 Catalina (Oct 7, 2019)

- Removed 32-bit apps
- *iTunes split into Music, TV, Podcasts*
- Sidecar (iPad as second display)

32. macOS X 11 Big Sur (Nov 12, 2020)

- *Biggest redesign (Design overhaul)*
- Apple Silicon (M1)
- New Control Center

33. Windows 11 (Oct 5, 2021)

- Centered Start Menu
- Rounded corners
- Snap Layouts
- Better gaming
- Android app support (later updates)
- *Sum...Major designs*

34. macOS 12 Monterey (Oct 25, 2021)

- Universal Control
- *AirPlay to Mac*
- Focus Mode

35. macOS 13 Ventura (Oct 24, 2022)

- *Stage Manager*
- Continuity Camera
- Passkeys

36. macOS 14 Sonoma (Sep 26, 2023)

- *Desktop Widgets*
- *Game Mode*
- Better video conferencing

37. macOS 15 Sequoia (Sep 16, 2024)

- Apple Intelligence (AI)
- *iPhone Mirroring*
- Improved productivity

38. macOS 16 Tahoe (Jun 29, 2026)

Q&A

1. Which Windows version first introduced the Start Menu and Taskbar? **Windows 95**
2. What does "XP" in Windows XP stand for? **eXPerience**
3. What company developed Mac OS? **Apple Inc.**

4. Why did Microsoft skip naming a version "Windows 9"? **Legacy code might confuse it with Windows 95/98**
5. MS-DOS stands for _____ **Microsoft Disk Operating System**
6. Which macOS version introduced Dark Mode? **macOS Mojave**
7. Even though it became "Windows", what was Bill Gates's original name for the system? **Interface Manager**
8. What version of Windows OS was Solitaire first introduced? **Windows 3**
9. Which Windows version removed the Start Menu and used the Metro UI? **Windows 8**
10. Which macOS version first introduced the Time Machine backup feature? **macOS Leopard**
11. Which macOS version broke the "big cat" naming trend and was named after a surf spot? **macOS Mavericks**
12. Which Windows version became infamous for being buggy and was nicknamed "Mistake Edition"? **Windows ME**
13. What do you call the Mac OS 10.16? **There is no Mac OS 10.16**
14. Which version of Microsoft Windows was the first to introduce the Start button? **Windows 95**
15. Arrange the following: Mavericks, Yosemite, El Capitan **Mavericks, Yosemite, El Capitan**
16. What is the latest version of MacOS? **MacOS 15 Sequoia**

Components of OS

- Process Management
- File Management
- Network Management
- Main Memory Management (RAM)
- Secondary Storage Management
- I/O Device Management
- Security Management
- Command Interpreter System (Shell)

Process Management

It handles the **creation, scheduling, termination, synchronization, communication and deadlock handling** of processes.

Process - a running or active program

File Management

It is responsible for **creating, deleting, reading, writing, and organizing and backing up files** on storage media.

Network Management

Network management **oversees communication between computers over a network**. It ensures data is transmitted between computers reliably and efficiently.

Main Memory Management

It handles the **allocation and deallocation of memory space to various processes**. This ensures that each process has enough memory to execute while optimizing the overall use of available memory. (RAM temporary data)

Secondary Storage Management

Responsible for **managing data storage on hard drives and other storage devices**. It offers secondary storage to back up the main memory. (SSD keeps data)

I/O Device Management

Responsible for **managing all input and output devices**, such as keyboards, mice, printers, monitors, and storage devices. The OS handles communication between the computer and these devices, ensuring efficient and error-free data transfer.

Security Management

Security management **protects the computer system from unauthorized access and ensures data integrity**.

Command Interpreter System

This is the part of the OS that interacts with the user, accepting commands and executing them (CLI or GUI). It acts as a bridge between user input and OS functions.

Summary:

Component	Function	Examples
1. Process Management	Handles program execution and multitasking	Task Manager, CPU scheduler, Running Apps
2. File Management	Organizes, stores, and retrieves files	File Explorer, Save/Open/Move/Delete files
3. Network Management	Manages data over the network	Wi-Fi settings, LAN connection
4. Main Memory Management	Allocates and frees RAM for programs	RAM allocation to apps, Memory leak prevention
5. Secondary Storage Management	Manages long-term storage	Hard Disk (HDD), Solid-State Drive (SSD), USB Drive
6. I/O Device Management	Controls input/output devices	Keyboard, Mouse, Monitor, Device Driver
7. Security Management	Protects system from threats and controls access	Password login, Antivirus, Firewall
8. Command Interpreter System (Shell)	Interface for user commands	Command Prompt (CMD), PowerShell

1. **Scenario:** You pressed a key on the keyboard and a letter appeared on screen.
 - **Component:** I/O Device Management
2. **Scenario:** Your computer asked for a password when logging in.
 - **Component:** Security Management
3. **Scenario:** You are streaming a video online through YouTube.
 - **Component:** Network Management
4. **Scenario:** You typed `cd Desktop` in the command prompt.
 - **Component:** Command Interpreter System (Shell)
5. **Scenario:** Your laptop lets you open PowerPoint, Spotify, and Chrome at the same time.
 - **Component:** Process Management
6. **Scenario:** You downloaded a large file and stored it on your hard drive.
 - **Component:** Secondary Storage Management
7. **Scenario:** Your printer started printing your file after you clicked "Print."
 - **Component:** I/O Device Management

Properties of an Operating System

MULTITASKING

Multitasking is the ability of an OS to run multiple applications at the same time.

HOW IT WORKS:

- The CPU time is divided into tiny slices.
- The OS uses a scheduler to switch between active tasks every few milliseconds.
- Each app gets a time slice to use the CPU.

MULTI-USER

A multi-user OS allows multiple people to use the same system—either simultaneously (like on a server) or one after another with separate profiles.

HOW IT WORKS:

- The OS maintains separate user accounts and permissions.
- It keeps each user's files, apps, and settings private.
- In servers, resource sharing is done in real-time with user isolation.

MULTITHREADING

- Multithreading is the ability of an OS to run different parts (threads) of the same program simultaneously.

HOW IT WORKS:

- Each program can be split into smaller tasks (threads) that run at the same time. Threads share the same memory, making programs faster and more efficient.

MULTIPROCESSING

Multiprocessing means the OS can use more than one processor (CPU or core) at the same time to complete tasks in parallel.

HOW IT WORKS:

- The OS splits tasks into chunks and sends them to different CPU cores.
- This makes execution faster and smoother, especially for demanding applications.

PORTABILITY

A portable OS can run on different types of hardware platforms with little or no change in its code.

HOW IT WORKS:

- Written in high-level languages (like C).
- Developers can compile the OS for different processors (Intel, ARM, etc.)

TIME-SHARING

Time-sharing means multiple users or tasks get a small time slot with the CPU so they can work "at the same time" from the user's perspective.

HOW IT WORKS:

- The OS gives each user or process a time quantum.
- It switches between them using round-robin scheduling.
- If one task takes too long, it is paused and the next gets CPU time.

DISTRIBUTED

A distributed OS manages a group of independent computers and makes them look like a single system to the user.

HOW IT WORKS:

- Uses networking to connect multiple systems.
- Tasks are distributed to different machines.

REAL-TIME

A Real-Time OS (RTOS) can respond to events within a guaranteed time frame—delays are not acceptable.

HOW IT WORKS:

- Prioritizes critical tasks (interrupts) over non-critical ones.
- Processes data as it comes, with low-latency response time.
- Often used in embedded systems where timing is crucial.

PROPERTIES OF AN OS:

PROPERTIES OF AN OPERATING SYSTEM	
Home	Content Contact
Property	What It Does
Multitasking	Run multiple apps at once
Multi-user	Multiple people use the system
Multithreading	One app runs multiple sub-tasks
Multiprocessing	Multiple CPUs work in parallel
Portability	Can run on many hardware types
Time-sharing	Fair use of CPU for many tasks
Distributed	Manages multiple computers as one
Real-time	Responds instantly to critical events

1. **Scenario:** A user opens Google Chrome, Microsoft Word, and Spotify all at once.
 - **Property:** Multitasking
2. **Scenario:** A Linux OS is installed and working smoothly on both a laptop and a Raspberry Pi.
 - **Property:** Portability
3. **Scenario:** An airbag system in a car responds instantly during a crash.
 - **Property:** Real-time
4. **Scenario:** A computer lab has different student accounts on the same machine.
 - **Property:** Multi-user
5. **Scenario:** A video editing program splits rendering tasks across multiple cores.
 - **Property:** Multiprocessing
6. **Scenario:** A chat application handles sending messages, receiving replies, and showing notifications.
 - **Property:** Multithreading
7. **Scenario:** The school server allows 30 students to compile their code at the same time.

- **Property:** Time-sharing

8. **Scenario:** Google Drive stores your files and lets you access them from any computer.

- **Property:** Distributed

9. **Scenario:** A phone OS handles a call while updating apps in the background.

- **Property:** Multitasking

10. **Scenario:** A drone instantly adjusts its propellers to balance in the wind.

- **Property:** Real-time